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Identifying the Need for New Knowledge

At first when we were trying to figure out what to do we weren't certain of anything at all regarding stocks or we had no baseline knowledge of the subject. We needed to know generally what kinds of things to put into the application.

Let's consider the example of us learning about candlesticks. Immediately into our cursory foray into stock trading we heard about candlesticks. None of us knew what they were or even what they were useful for. We then assigned people the task of researching what candlesticks were and explain the findings to the group in a presentation of a kind. We were tasked with answering those two questions of “what?” and “why?”. Nathan then used google to research the topic and figured out what they were and what they were useful for, namely that they were open, close, high, and low over a given time period. And they were useful because depending on their shape some investors could guess what will happen with the stock.

We primarily used online articles, trading journals, other stock monitoring apps, and even some videos to figure out what was going on.This was a typical method for research among all of the members and the biggest differences in approaches came with what domain we were researching. If it was a question about the business domain we looked at articles and posts, if it was a technical question we referred to the documentation and asked AI. After we had a base level understanding we wanted to know what specific candlestick patterns would be the most useful so we assigned Chris to compose a list of specific candlestick patterns that might be useful for patterning. At this moment in the project we were thinking this information might be useful in modeling and that we might try and see what stocks match up with what patterns and predict them that way. As we learned more about the patterns themselves we saw them as interesting but not necessarily trustworthy.

Applying the Knowledge

The next thing that we did with this task was try to apply this knowledge to the project. We thought that these candlestick patterns were interesting enough pieces of information that it would be good showing them to the user if a given stock matched that pattern. We then gave a task to Logan to create an endpoint in our api to display what if any candlestick patterns a given stock matched up for as a proof of concept. The proof of concept went well enough that we felt comfortable adding that in as a metric in the database. This opened it up to being filterable and visible on the stocks display page.

The information that we had gathered proved invaluable when designing features for our app. It was specifically useful with filters and filtering and while the initial plans that we had for that research changed it was still important enough that we used it in other parts of the app.

Reflection on the Process

The learning strategy is on the whole was effective. We would definitely use this same strategy in future projects with some of the specifics refined. You really don’t have to reinvent the wheel with research. Just stick with the basic strategy of: research a topic broadly, pick a specific aspect of that topic and research that, then try to apply it. It is simple and works generally pretty well. Also be smart about what domain that you are trying to research. If you have a technical question go to the technical documents if you have a business domain question go to the people in that business.

Nathan reflecting on this process said “If I had to change or modify that approach at all I would probably use more Wikipedia. Wikipedia is a great resource to get broad data and specific information that you need with decent documentation and citing of sources. If you want to dive further after looking at a Wikipedia article you are easily able to by going to its citations.”

Chris said “For me, personally, the next time I start a project like this, I would definitely do a lot more research and experimentation with API's and see the ups and downs of using each. I think if we were using polygon from the beginning, we would have had more time to add the other features we wanted “

Logan said “I would look to spend more time inside of the documentation when researching programing related tasks. I relied a lot on generated code when researching this project which I could work to understand better. Overall spending more time understanding the pros and cons of an approach would have gone a long way.”